Credit Card Financial Dashboard Analysis

1. Introduction

The Credit Card Financial Dashboard project focuses on **analyzing credit card transaction and customer data to monitor weekly performance and financial trends**. The organization generates **large volumes of transaction data every week**, making manual analysis time-consuming and inefficient. This project aims to transform raw SQL data into meaningful insights using Power BI.The dashboard enables stakeholders to monitor performance in real time and make data-driven decisions efficiently.

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2. Objective of the Project

The objectives of this project are:

- To analyze credit card transaction and customer data.
- To monitor weekly revenue and transaction trends.
- To reduce manual reporting effort.
- To track key financial performance metrics.
- To support data-driven decision making.

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3. Problem Statement

Previously, credit card performance analysis was not streamlined, leading to several challenges:

- Manual preparation of weekly reports was time-consuming.
- Difficulty in tracking week-over-week (WoW) changes.
- Limited visibility into customer segmentation and revenue contribution.
- Data stored in SQL database was not fully utilized for analytical reporting.

Therefore, a centralized and automated dashboard solution was required to generate quick and actionable insights.

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4. Tools Used

The following tools and technologies were used:

- **Power BI** – Dashboard creation and visualization
- **SQL Workbench** – Data extraction from database

- **MySQL Database** – Source of transaction and customer data

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5. Methodology

The project was completed using the following steps:

1. Extracted data from the CSV files to MySQL database using SQL-Queries
2. Imported data into Power BI.
3. Cleaned data by removing invalid and duplicate records.
4. Created calculated columns (Age Group, Income Group).
5. Developed DAX measures such as:

5.1. Revenue

5.2. Current Week Revenue

5.3. Previous Week Revenue

5.4. Week Number

6. Designed interactive visuals and KPI cards.
7. Designed interactive dashboard visuals.

Figure 1: Credit Card Financial Transaction Dashboard Report-01

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6. Dashboard Insights

The dashboard provides insights such as:

Week 53 (31st December) – WoW Change :

- Revenue increased by **28.8%**
- Total transaction amount increased
- Total transaction count increased

- Customer count increased

<u>Overall Year-to-Date (YTD) Insights</u> :

- Overall Revenue: **57M**
- Total Interest Earned: **8M**
- Total Transaction Amount: **46M**
- Male customers contributed **31M revenue**
- Female customers contributed **26M revenue**
- Blue & Silver credit cards contributed to **93% of total transactions**
- TX, NY & CA contributed to **68% of overall revenue**
- Overall Activation Rate: **57.5%**
- Overall Delinquency Rate: **6.06%**

These insights help stakeholders identify revenue drivers, customer contribution patterns, and financial risks.

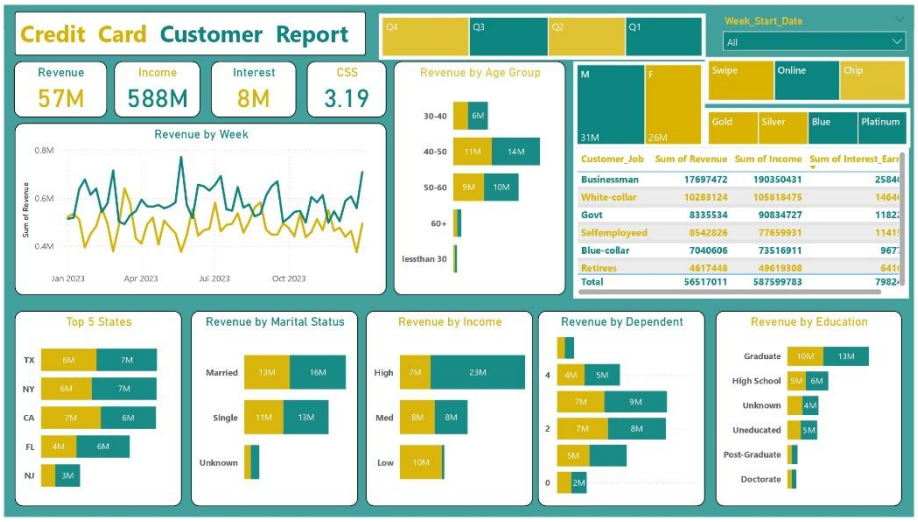

Figure 2: Credit Card Financial Customer Dashboard Report-02

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7. <u>Outcome of the Project</u>

The project resulted in:

- An automated and interactive weekly financial dashboard
- Real-time tracking of revenue and transaction trends
- Improved monitoring of activation and delinquency rates
- Faster decision-making through visual insights
- Clear visibility into customer segmentation and state-wise contribution